GOVERNMENT OF KARNATAKA

Department of Collegiate and Technical Education

Government Polytechnic for Women – Bengaluru

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that Kumari …….. student bearing the Reg No: …….. of 4th semester Diploma in Computer Science and Engineering has successfully completed the Object Oriented Programming And Design with Java Activity prescribed by the Board of Technical Education, Bengaluru during the academic year 2025-2026 (EVEN).

Signature of the Course Coordinator Signature of the Program Coordinator

GOVERNMENT OF KARNATAKA

Department of Collegiate and Technical Education

Government Polytechnic for Women,Bangalore

Department of Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN WITH JAVA (20CS43P)

ACTIVITY

Submitted by

Name :

Reg No :

4TH Sem CSE

Course Co-ordinator
Mrs. REKHA A.NAIK

LECTURER

Department of Computer Science and Engineering
2025-2026

INDEX

SL No	WEEK	ACTIVITY	Submission Date	EVALUATION	
				Marks	Signature
1	1	a) Identify various java IDEs and identify differences between them. b) Compare and contrast Java with Python			
2	2	1. Study and present a). type casting in java b). what are command lines arguments in java? c). java keywords and their usage			
3	3	Compare and contrast a). method and constructor; b). constructor and destructor			
4	4	a) Study and present how does bytecode work in java			
5	5	a). Present nesting of conditional and iterative statements considering a use case.			
6	6	Identify advantages and disadvantages of a. Encapsulation. b. Inheritance c. Abstraction d. Polymorphism			
7	7	Study and report a. java Arrays class their methods b. java String class their methods			

8	8	Identify and document how these principles help to avoid code smells. a. SRP b. OCP c. ISP			
9	9	1. Code, execute and debug programs that uses a. static binding b. dynamic binding			
10	10	1. Code, execute and debug programs that uses 2. abstract class to achieve abstraction 3. interface to achieve abstraction 4. Verify whether the given code snippet is correct according to abstraction or not			
11	11	a) Study DRY principle, identify the benefits.			
12	12	Identify how OOD principles violations impact the quality of code.			
13	13	Identify java ORM frameworks and their features.			

GOVERNMENT POLYTECHNIC FOR WOMEN Bengaluru -560001

ACTIVITY EVALUATION SHEET : 2025-26			
NAME:		REG.NO:	
PROGRAMME	COMPUTER SCIENCE AND ENGINEERING	SEM :	4TH SEM
COURSE :	OOP AND DESIGN WITH JAVA	COURSE CODE :	20CS43P
WEEK :	1-13	MAX MARKS :	10

	Scale					
Dimension	**2 Beginners**	**4 Intermediate**	**6 Good**	**8 Advanced**	**10 Expert**	**Marks**
1.focus on the task and gathering information	Does not collect information related to topic	Collects very limited information, some relate to topic	Collects basic information , most refer to the topic	Collects more information, most refer to the topic	Collects a great deal of information, all refer to the topic	
2. Communication	Loses audience's attention due to lack of clarity	Unclear, difficult to understand	Audience must put forth effort to listen, poor pronunciatio n	Occasionally difficult to understand	Clear, easy to understand	
3. presentation	Tension and nervousness are obvious, multiple mistakes	Tense body language, frequent mistakes	Some tension noted, stiff body language	Occasional mistakes but recovers quickly	Prepared the report and	
4. Prepare the report and time manageme nt	Never prepared the report and not submitted	Prepared the report and not submitted	Prepared the report and Take too much time to submitted	Prepared the report and late submitted	Timely submitted	

Signature of the Student

Signature of course Coordinator